

Machine Learning Assignment Week 3

Dataset Header: #1-1-1

Stephen Haskins: 21365534

October 15, 2025

Introduction

A)

- i) Visualise the data on a 3D scatter plot
- ii) Train a Lasso Regression model and Predict target values
- iii) Plot the results
- iv) Explain underfitting and overfitting
- v) Repeat for Ridge Regression

B) Support Vector Machines

- i) Use 5-fold cross-validation to determine the best value of C
- ii) What value of C is optimal?
- iii) Repeat for Ridge Regression

Conclusion

References

Appendix: Code